

7. Operation mode

After the answer to reset, the card is in one of the following two operating modes:

- Dedicated mode when TA2 exists;
- Negotiation mode when TA2 is not present.

7.1 Dedicated mode

In the dedicated mode, immediately after the answer to reset, the protocol indicated by TA(2) is used.

- When b5=0 in TA(2), Fi and Di are used. After the reset response, Fi and Di take effect immediately.
- When b5=1 in TA(2), the default value (Fd, Dd) is used.

7.2 Negotiation Mode

In the negotiation mode, as long as the first byte sent by the IFD to the ICC allows a significant difference between the PPS request and the protocol command, the "default selection" is possible.

- If there is no PPS request after the answer to reset, the "preferred protocol" will use Fd and Dd (default values).
- When the protocol is provided by ICC and/or other values of parameters F and D (F range from Fd to Fi, D range from Dd to Di), IFD should send a PPS request with Fd and Dd to change from negotiation mode Switch to dedicated mode, after successfully completing the PPS exchange, the negotiation protocol should use Fn and Dn (defined in PPS1).

If the answer to reset only provides one protocol (T=0 to 14) and Fd, Dd, the protocol should use Fd and Dd and follow the answer to reset. Accordingly, such a card does not need to support PPS.

An IFD that neither supports PPS nor the preferred protocol can reset the ICC to switch from the negotiation mode to the dedicated mode supported by the IFD, or the card can be rejected.

If T=0 or T=1 with values Fi and Di and Fi and Di are not equal to Fd, Dd, then IFD can choose the default protocol with Fd and Dd, or send a PPS request with Fd and Dd to coordinate Fn , Dn.

8. PPS protocol in negotiation mode

Protocol and Parameters Selection, literally translated as protocol and parameter selection, is a process in which the reader and the card negotiate the communication protocol and transmission parameters. It has two modes of operation: a dedicated mode when TA(2) exists and a negotiation mode when TA(2) does not exist.

1. Dedicated mode: Fi and Di are used when bit5=0 of TA(2), and the default value is used when bit5=1.

2. Negotiation mode: As long as the first byte of IFD to ICC allows a significant difference between the PPS request and the protocol command, the "default selection" is possible.

There is no PPS request after the answer to reset, the "preferred protocol" will use Fd and Dd.

When the protocol is provided by ICC and/or other values of the parameters F and D (the range of F is Fd to Fi, and the range of D is Dd to Di), IFD shall send a PPS request with Fd and Dd in order to change from negotiation mode Go to dedicated mode. After successfully completing the PPS exchange, the negotiation protocol should use Fd and

Dd.

Only IFD is allowed to start PPS exchange

- IFD should send a PPS request to ICC;
- If the ICC receives an incorrect PPS request, it does not respond;
- If the ICC receives a correct PPS request, it should return a PPS response otherwise it will exceed the initial waiting time;
- If the initial waiting time is exceeded, the IFD will either reset or reject the IC card;
- If the IFD receives a wrong PPS response, the IFD either resets or rejects the IC card;
- If the PPS exchange fails, the IFD either resets or rejects the IC card.

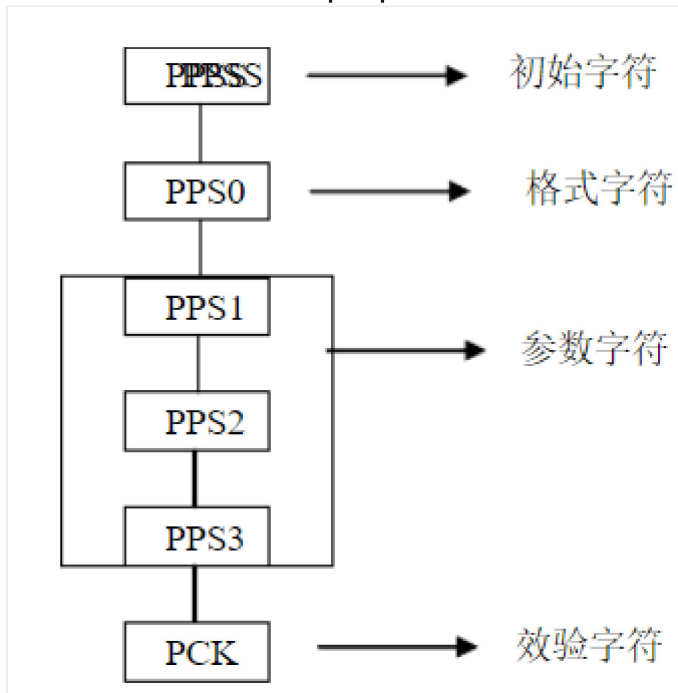
8.1 The structure and content of the PPS request

PPS request and response respectively include an initial byte PPSS followed by format byte PPS0, three optional parameter bytes PPS1, PPS2 and PPS3, and a detection byte PCK.

PPS identifies PPS request or response and is equal to FF.

PPS0 indicates the existence of optional bytes PPS1, PPS2, and PPS3 through bits b5, b6, and b7. Bits b4~b1 transmit the value of parameter T to propose a protocol. Bit b8 is reserved for future use and set to 0.

PPS1 allows IFD to propose F and D values to the card.



8.2 Successful PPS exchange

When the PPS response is one of the following, the PPS exchange is also successful.

(1) PPSS response = PPSS request;

(2) PPS0 response:

- B1 to B4 should be returned
- Send back B5 or set it to 0 (if B5=1, PPS1 response=PPS1 request. If B5=0, no PPS1, use Fd and Dd).
- Send back B6 or set it to 0 (if B6=1, PPS2 response=PPS2 request. If B6=0, PPS2 request and corresponding do not exist).
- Send back B7 or set it to 0 (if B7=1, PPS3 response=PPS3 request. If B7=0, PPS3 request and corresponding do not exist).

All other cases of PPS exchange should be interpreted as unsuccessful.

The data details of the PPS response are the same as the details of the PPS request, but the settings need to comply with the following rules:

Bits 1~4 in response to PPS0 are the same as bits 1~4 in PPS request.

Bits 5~7 of response PPS0 are the same as bits 5~7 of request PPS, or set to 0.

Bit8 in response to PPS0 is also reserved.

In fact, most of the time, responding to PPS is the same as requesting PPS.

8.3 PPS negotiation example

If the PPS response accurately reflects the PPS request, then the PPS exchange is successful. That is to say, what the card reader sends to the card, the card returns to the reader, of course, the premise is that the PPS sent by the card reader is correct. As shown in the figure below, the blue character is sent to the card by the card reader, "FF 11 11 FF", where the first 'FF' is PPSS; the first '11' is PPS0, the upper 4 digits of 1 indicate PPS1 Exist, the lower 4 bits indicate T=1; the second '11' is PPS1, and Fi=1 and Di=1 are specified. It can be seen from the table that this is 372 frequency division; the last 'FF' is the first three XOR value of value. After that, the card sends the same data (white characters) as a reply, indicating that the PPS negotiation is successful. This is the most common situation and it is also a frequently used situation. Other situations may also occur.

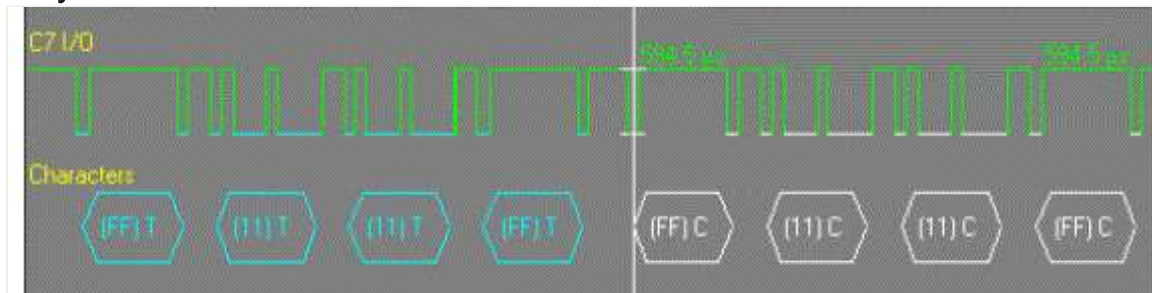


图 8 成功的 PPS 交换

Using Jinsite ElectronicsKingst LA5016 usb logic analyzerCheck IOS7816 data communication. The following pictures are screenshots of a complete PPS packet analysis. As can be seen from the figure, in the PPS request of the protocol, it can be checked that only PPS1 exists by parsing PPS0, and the PPS response is the same as the PPS request, which is also within the scope allowed by the standard.

Protocol analysis parameter settings:

